

PROBLEM STATEMENT

With the aggravation of the global greenhouse effect and environmental pollution, energy saving and emission reduction have already become the consensus of the manufacturing industry to enhance sustainability.

Automated Guided vehicles(AGVs) are increasingly utilized in various industries for material handling and logistics. Energy consumption is a key challenge, with an AGV typically using 0.22 kWh/ton-km per day. Reducing energy consumption not only benefits the environment but also lowers operational costs by minimizing battery usage and extending vehicle lifespan.

Path planning in AGV integrates energy consumption with traditional algorithms with goals like minimizing distance and time, using energy models and constraints to optimize both efficiency and cost

OBJECTIVE FUNCTION AND DESIGN VECTOR

- Minimize Distance Traveled (D) : finds the shortest path from the start point to the goal while navigating through the environment. This reduces travel time and increases efficiency.
- Minimize Energy Consumption (E): Energy efficiency is crucial for operational cost savings
- Efficient battery management for improving utilization

$$\min X = w_1 D + w_2 E$$

D = Total distance travelled by AGV

E = Total energy consumption

W1, w2 = weighing constants(Costs)

$$X = [dtotal, v, a, rturn, Edrive, Eturn, Eacc, Estandby, Pdrive, Pcontrol, Pload, Wload, g(n), h(n), f(n), Eunit]$$

Multi-objective problem

CONSTRAINTS

Design Variable	Symbol	Description
Speed	$V(t)$	Velocity of AGV at time t
Acceleration	$a(t)$	Acceleration of AGV at time t
Load Weight	W_{load}	Weight of transported load
Turning Angle	$\theta(t)$	Turning angle per movement step
Path Position	$x(t), y(t)$	AGV's position in the grid at time t
Grid Boundary	-	Constraints on the AGV's movement within the grid

Constraints

$$0.1 \leq a(t) \leq 0.5$$

$$0.5 \leq V(t) \leq 3.0$$

$$W_{load} \leq W_{max}$$

$$R_{turn}(t) \geq R_{min} = \frac{L}{\tan(\theta(t))}$$

$$(f'(n) = f(n) + \lambda(1 - \frac{\text{Battery Level}}{\text{Threshold}}))$$



MATHEMATICAL FORMULATION

Final A* -energy optimized model -

$$f(n) = g(n) + h(n)$$
$$= \sum_{i=1}^k (E_{\text{drive},i} + E_{\text{control},i} + E_{\text{load},i}) + d(n, \text{goal}) \times E_{\text{unit}}$$

Objective function -

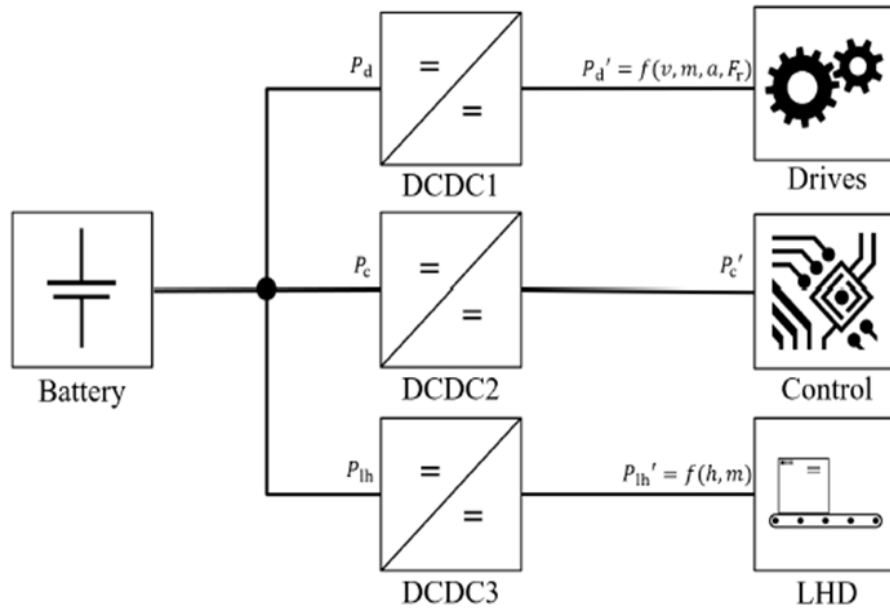
$$\min X = w_1 D + w_2 E$$

$$P_{\text{drive}}(t) = k_1 V(t) + k_2 a(t) + k_3 W_{\text{load}} V(t)$$

$$P_{\text{control}}(t) = \begin{cases} P_{\text{active}} & \text{if turning or navigating} \\ P_{\text{standby}} & \text{if idle or moving straight} \end{cases}$$

$$P_{\text{load}}(t) = \begin{cases} P_{\text{active-load}} & \text{if load handling event} \\ 0 & \text{otherwise} \end{cases}$$

LITERATURE REVIEW AND KEY REFERENCE



Component analysis : To model an ERM, components of AGV are defined first. Next Power requirement of the components are defined as a function of it's influencing factors. Finally the total energy requirement is the power requirement of components summed up over time.

Drives

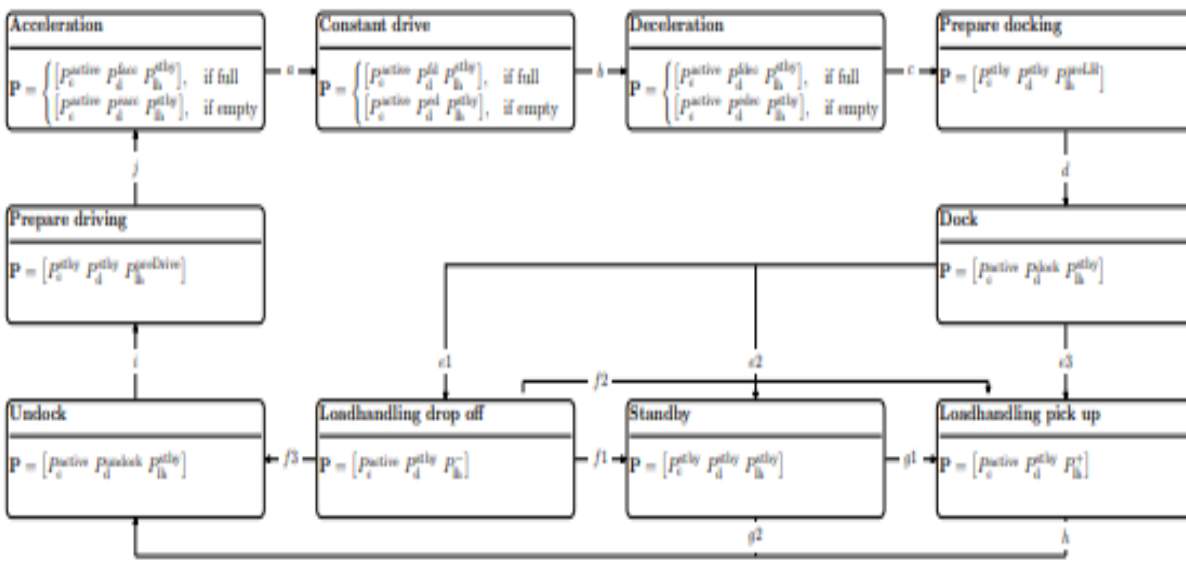
Electric power of the drives and motor controls as a function of velocity, acceleration

Control

Components responsible for control, communication, navigation, localization, and safety

LHD

Components that are involved in load handling process



Process and impact analysis :

Analysis to identify the process states of AGVs. The analysis reveals 10 different states

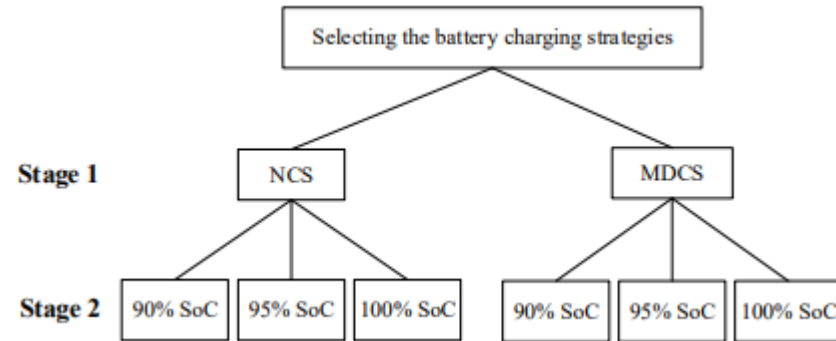
$$P = [P_{ctrl} \quad P_{drives} \quad P_{lh}]$$

1. Acceleration
2. Constant drive
3. Deceleration
4. Prepare docking
5. Dock

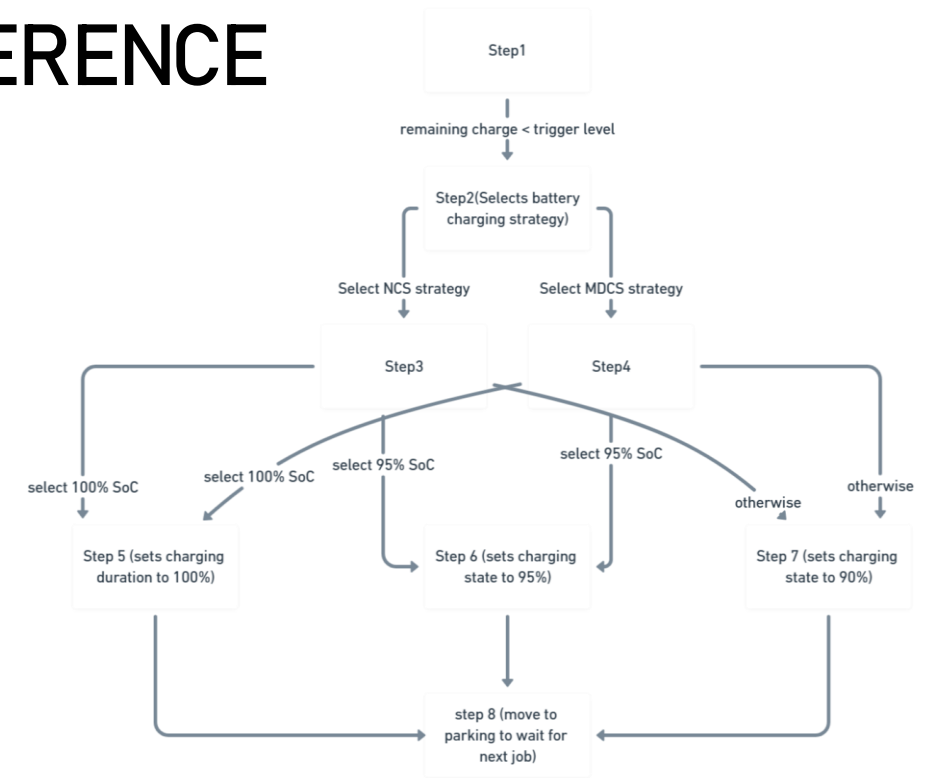
6. Load handling (pick up and drop off)
7. Standby
8. Prepare driving
9. Undock

LITERATURE REVIEW AND KEY REFERENCE

Study on AGVs battery charging strategy for improving utilization



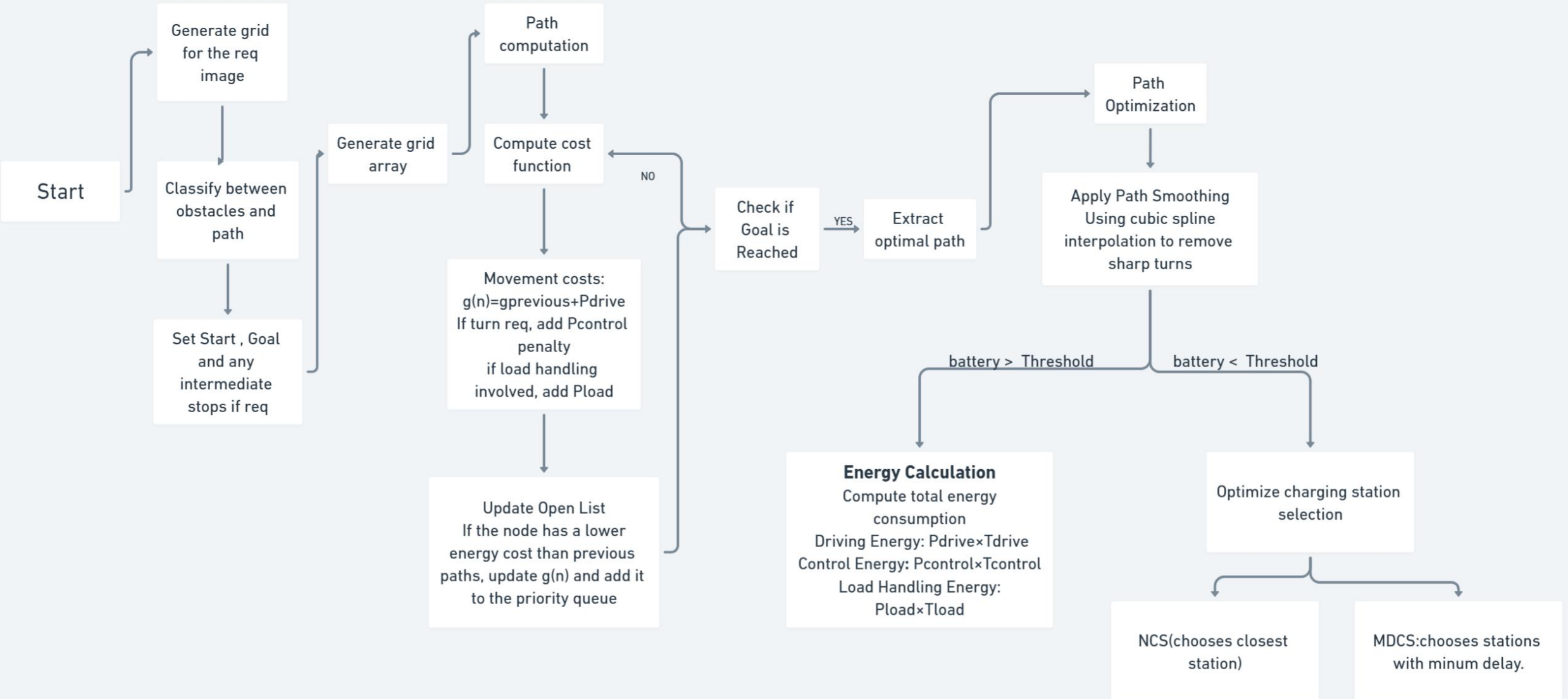
$$E_{\text{state},i} = \int_{t_0}^{t_0+t_{\text{state},i}} P(t) dt, \quad i \in I$$



The total scalar energy requirement $E(t)$ is the sum of all individual states as a function with the evaluation period t , and the dispatching method used . Mathematically, formulated follows:

$$E = \sum_{i \in I} \sum_{k=1}^3 E_{\text{state},i,k}$$

METHODOLOGY



ALGORITHM USED -

Standard A* cost function

$g(n)$ = actual cost from start to node n

$h(n)$ = heuristic estimated cost from n to the goal.

$$f(n) = g(n) + h(n)$$

Energy-Based Cost Function (Modified $g(n)$):

$E_{\text{state},i}$ = energy cost in each state (acceleration, turning, standby)

K = the number of steps taken so far

$$g(n) = \sum_{i=1}^k (E_{\text{state},i})$$

Using the ERM equation from paper :

Driving energy : P_{drive} = power consumption of the AGV's motors.

$$E_{\text{total}} = E_{\text{drive}} + E_{\text{control}} + E_{\text{load-handling}}$$

$$E_{\text{drive}} = P_{\text{drive}} \times t_{\text{drive}}$$

Control system energy : P_{control} = control power usage.

$$E_{\text{control}} = P_{\text{control}} \times t_{\text{active}}$$

Load-handling energy: P_{load} = power consumption for material handling

$$E_{\text{load}} = P_{\text{load}} \times t_{\text{handling}}$$

Energy-Aware Heuristic $h(n)$ -

$d(n, \text{goal})$ = estimated remaining distance.

E_{unit} = average energy per unit distance (calculated from dataset).

$$h(n) = d(n, \text{goal}) \times E_{\text{unit}}$$

ALGORITHM USED -

Final A* -energy optimized model -

$$f(n) = g(n) + h(n)$$

$$= \sum_{i=1}^k (E_{\text{drive},i} + E_{\text{control},i} + E_{\text{load},i}) + d(n, \text{goal}) \times E_{\text{unit}}$$

Define Energy Costs for Movements-

Instead of using distance-based weights, we assign energy consumption values from the dataset to different AGV actions.

Moving straight → Energy demand drives

Turning → Energy demand control

Waiting/standby → T standby data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Experiment	Versuch-ID	Hour-ID	Simulation results			Experimental results			Energy demand control		Energy demand drives		Energy demand Loadhandling		
2				E.d. Contr	E.d. Drives	E.d. Loadh	E.d. Control	E.d. Drives	E.d. Loadhandlin	absolute deviation	Relative deviation	absolute d	Relative deviation	absolute d	Relative deviation	Simulation
3	anta Setup13	Versuch1	1	61.477140	24.4262797	14.0743219	62.10794366	25.51490959	14.59019599	-0.630804	-1.02%	-1.088630	-4.27%	-0.515874	-3.54%	99.97
4			2	61.307832	24.1005347	13.6983457	61.92265806	24.76780852	14.0600935	-0.614826	-0.99%	-0.667274	-2.69%	-0.361748	-2.57%	99.1067
5			3	61.294431	24.0670632	13.898632	61.82110628	24.09341896	14.08761717	-0.526676	-0.85%	-0.026356	-0.11%	-0.188985	-1.34%	99.2601
6			4	61.257576	23.9460846	14.1240267	62.10564643	24.11816552	14.11101706	-0.848070	-1.37%	-0.172081	-0.71%	0.013010	0.09%	99.32
7			5	61.602446	24.774465	14.2256764	61.94144624	24.84132679	14.60799216	-0.339001	-0.55%	-0.066862	-0.27%	-0.382316	-2.62%	100.60
8			6	61.294431	24.005013	13.891128	61.69579699	24.07213622	14.17550826	-0.401366	-0.65%	-0.067123	-0.28%	-0.284380	-2.01%	99.1905
9			7	61.294431	24.005013	13.891128	61.68015227	24.20524842	14.16844876	-0.385722	-0.63%	-0.200235	-0.83%	-0.277321	-1.96%	99.1905
10			8	61.29443056	24.0215884	13.8932124	61.65709961	24.06415817	14.23749047	-0.362669	-0.59%	-0.042570	-0.18%	-0.344278	-2.42%	99.2092
11			9	61.565591	24.7078902	14.4585751	61.5511882	24.3420544	14.32664457	0.014403	0.02%	0.365836	1.50%	0.131931	0.92%	100.732
12			10	61.307609	24.0999131	13.8137528	61.65955657	24.87446416	14.80947538	-0.351948	-0.57%	-0.774551	-3.11%	-0.995723	-6.72%	99.2212
13			11	61.300908	24.0950098	13.8530961	61.50268185	24.1371231	14.08744366	-0.201774	-0.33%	-0.042113	-0.17%	-0.234348	-1.66%	99.2490
14			12	61.294431	24.0499873	13.898632	61.61798701	24.36958395	14.30342568	-0.323556	-0.53%	-0.319597	-1.31%	-0.404794	-2.83%	99.2430

ALGORITHM USED -

Modifying the cost function in A*.

- Instead of distance, use energy consumption.
- Penalize high-speed acceleration to reduce Energy demand drives.
- Prefer paths with lower control energy costs (Energy demand control).

- **Straight movement cost** - default energy per step
- **Turning cost** - higher energy for changing direction
- **Acceleration penalties**
- **Standby energy cost**- if AGV needs to wait
- **Turning cost** - higher energy for changing direction
- **Acceleration penalties**
- **Standby energy cost**- if AGV needs to wait

Implementation

We get the energy consumption values from the dataset

Driving Energy:

Mean Power: 37.90 W
Standard Deviation: 6.85 W
Mean Drive Time: 33.70 s

Control System Energy:

Standby Power: 55.15 W
Active Control Power:
63.19 W

Load Handling Energy:

Standby Power: 7.50 W
Active Load Power: 54.71 W
Mean Load Handling Time: 8.67 s

ALGORITHM PARAMETERS FOR CONVERGENCE-



Parameter	Description	Effect on Convergence
Speed Limit	Maximum allowed speed for movement	Prevents unrealistic speeds and optimizes battery usage
Acceleration/Braking Cost	Limits how fast an agent can speed up or slow down	Reduces energy waste from sudden speed changes
Battery Constraints	Minimum energy required to reach the goal	If energy runs out before a charging station, the algorithm fails
Path Smoothing Penalty	Cost added for sharp turns or zigzag paths	Encourages efficient and direct paths
Charging Station Selection	Determines when and where to recharge	If energy runs out before a charging station, the algorithm fails
Energy Cost Model	The energy required to move to the next node	Ensures feasible paths under energy constraints



COMPLEXITY OF MODEL-

Time Complexity: $O(b^d)$

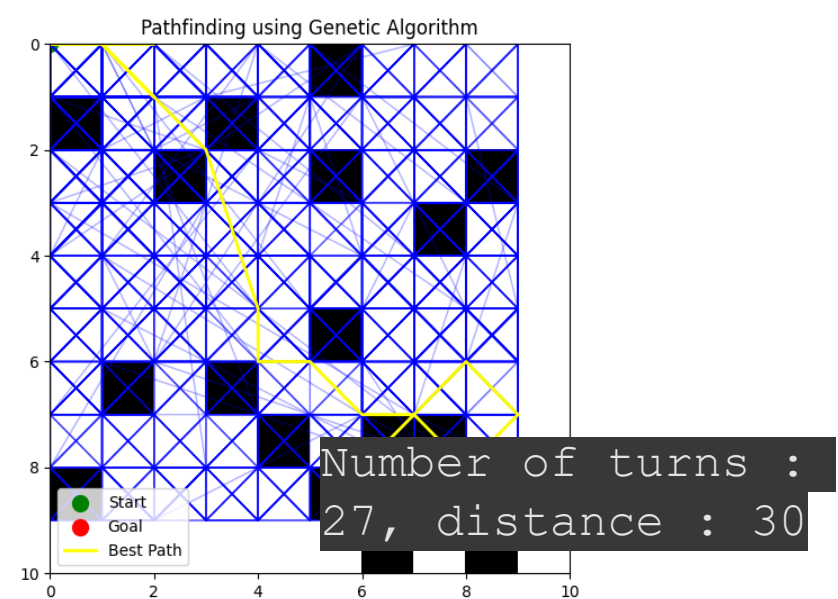
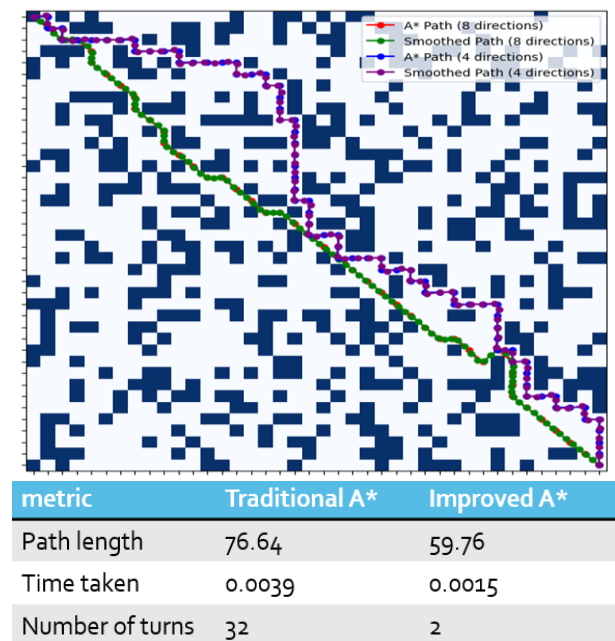
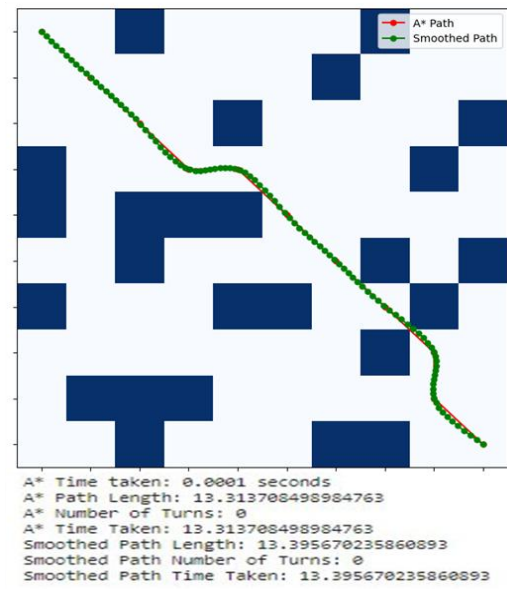
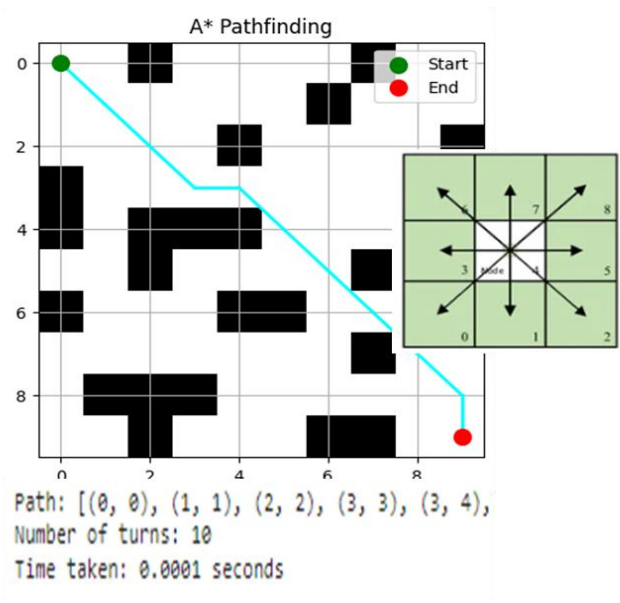
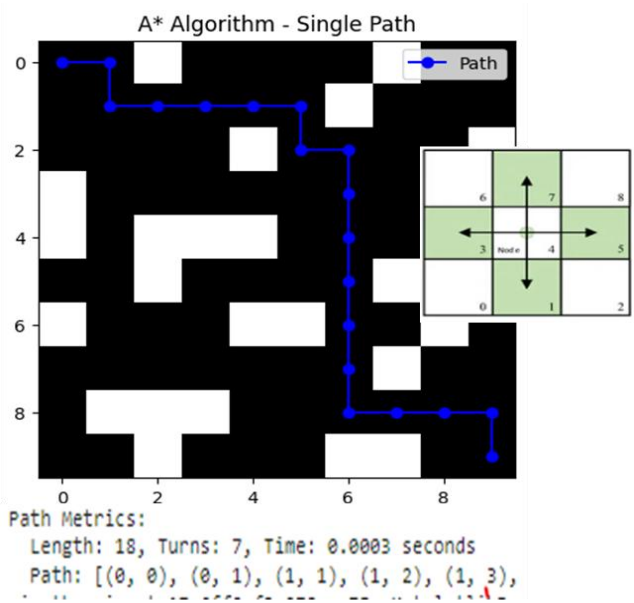
- **b** = branching factor (≈ 8 due to 8 directions)
- **d** = depth of the shortest path to the goal or charging station

Space Complexity: $O(b^d)$

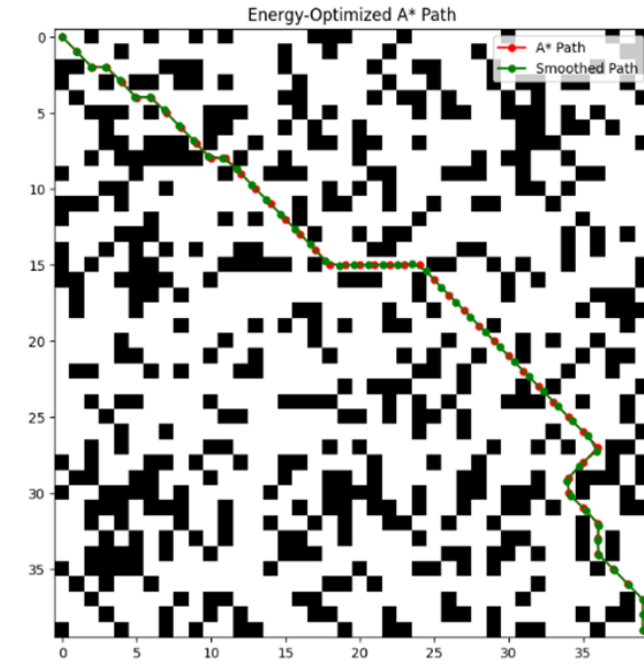
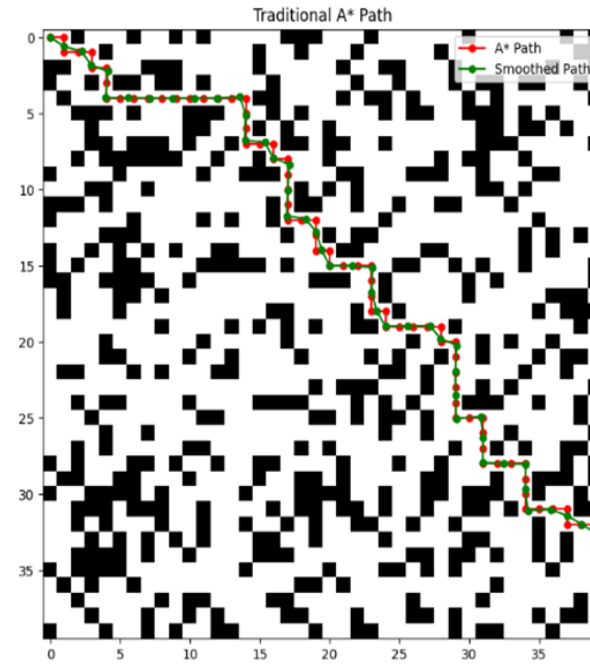
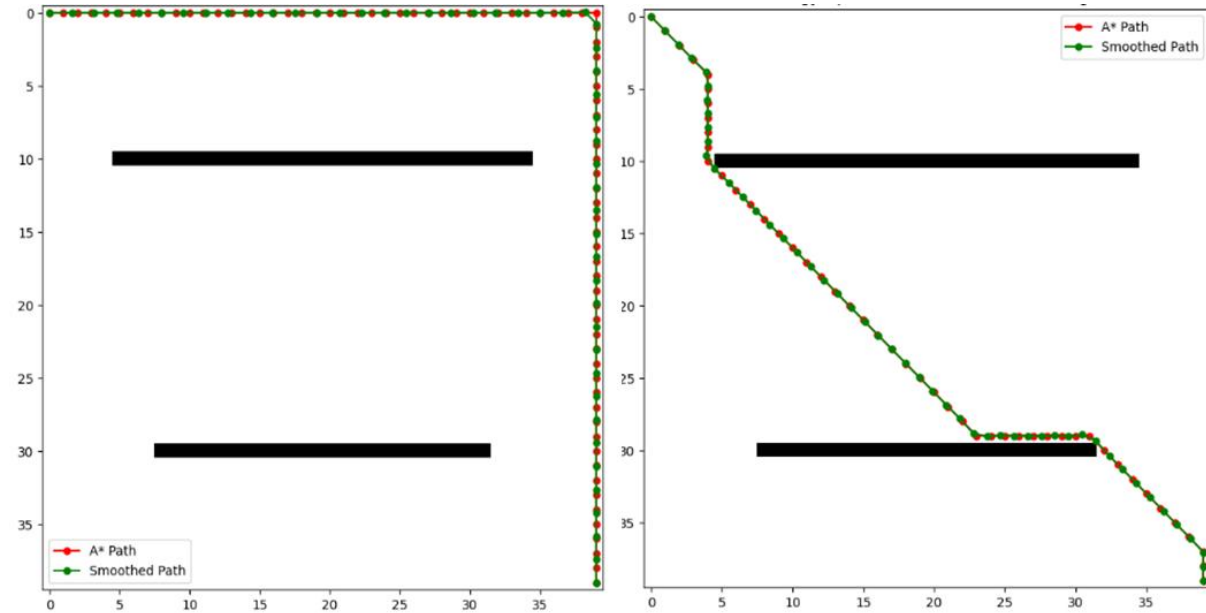
- g_score and f_score dictionaries $\rightarrow$ up to one entry per node
- came_from map
- The open list (heap)

Metric	Value
Time Complexity	$O(b^d)$
Space Complexity	$O(b^d)$
Branching Factor (b)	~ 8
Depth (d)	Grid-dependent

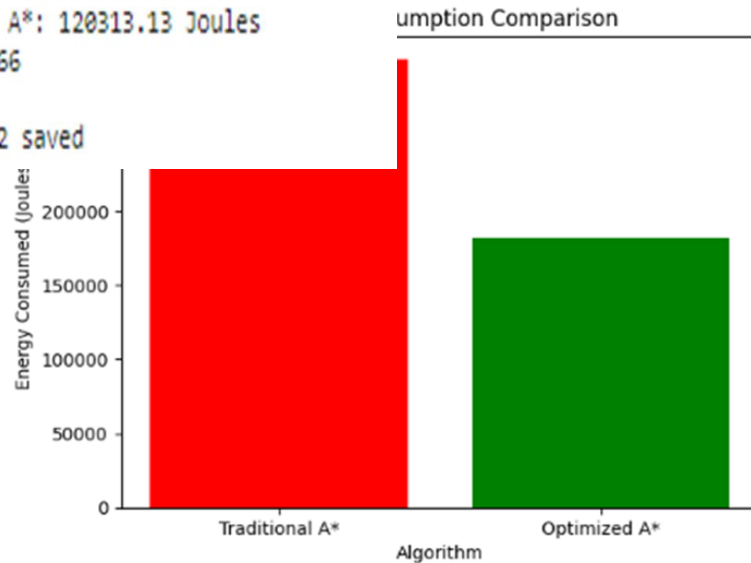
RESULTS



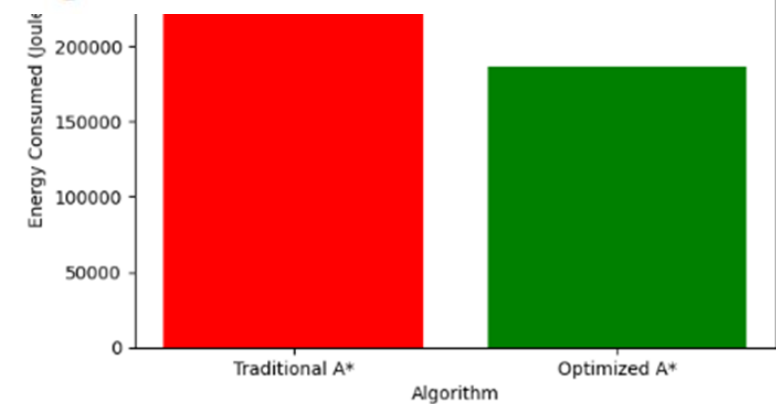
RESULTS



Energy Consumed by Optimized A*: 182410.23 Joules
 Energy Consumed by Traditional A*: 302723.36 Joules
 Energy Savings by Using Optimized A*: 120313.13 Joules
 Energy Efficiency Ratio (EER): 1.66
 Percentage Energy Savings: 39.74%
 Environmental Impact: 26.74 kg CO₂ saved



Energy Consumed by Optimized A*: 186291.30 Joules
 Energy Consumed by Traditional A*: 302723.36 Joules
 Energy Savings by Using Optimized A*: 116432.06 Joules
 Energy Efficiency Ratio (EER): 1.62
 Percentage Energy Savings: 38.46%
 Environmental Impact: 25.87 kg CO₂ saved



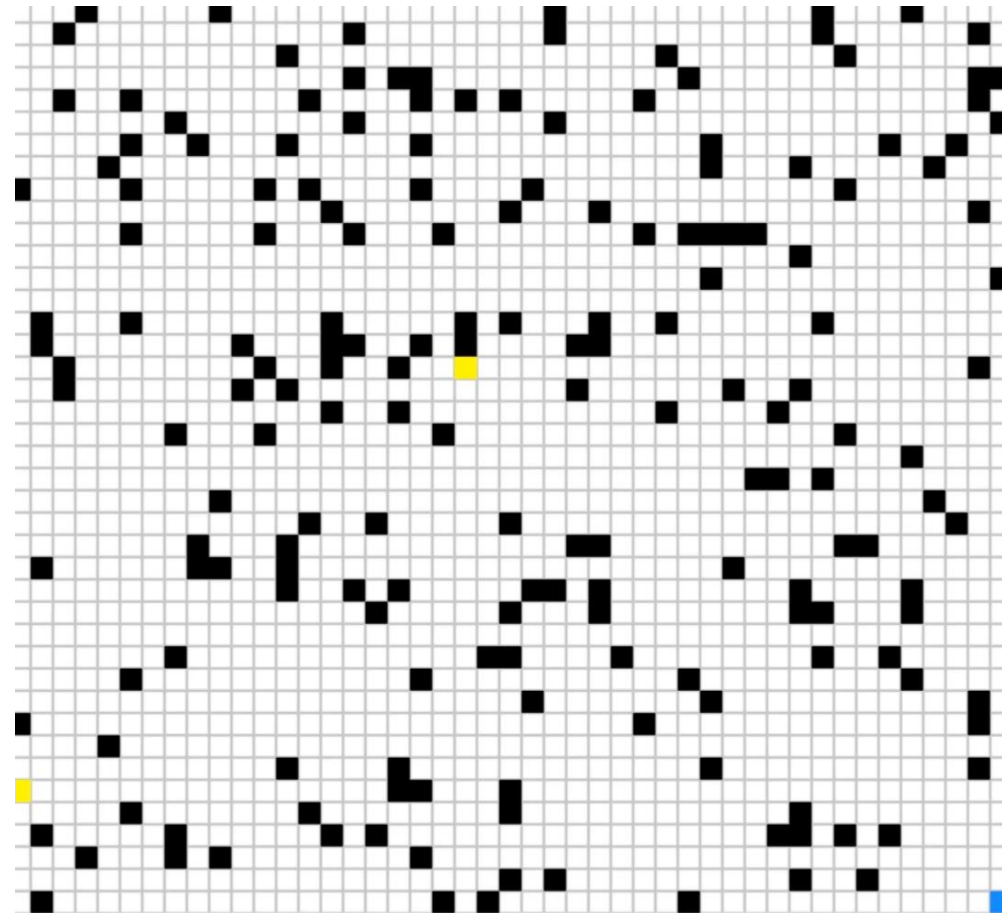
CODE AND DEMONSTRATION

[Improved_Astar_code](#)

[charging_stations_code](#)

[Genetic_algo_code](#)

[Visualizing_path](#)



Press [O] Optimized | [T] Traditional | [Q] Quit

SUMMARY

This AGV simulation project effectively models the navigation and energy management of Automated Guided Vehicles in a grid environment. By implementing both Energy-Optimized and Traditional A* algorithms, the simulation not only finds the shortest paths but also considers battery levels and energy consumption, which is crucial for real-world applications.

Applications

Warehouse Management: The simulation can be applied in warehouses to optimize the movement of AGVs, ensuring efficient routing while minimizing energy costs.

Manufacturing: In manufacturing settings, AGVs can be programmed to transport materials while managing battery life effectively, reducing downtime.

Benefits Cost Efficiency: By optimizing energy usage, organizations can lower operational costs associated with charging and maintenance.

Sustainability: The project highlights the environmental impact of AGV operations, allowing companies to make informed decisions that reduce their carbon footprint.

Scalability: The grid and obstacle configurations can be adjusted to simulate various environments, making the tool adaptable for different operational scenarios.

In summary, this project provides valuable insights into AGV behavior, helping organizations enhance efficiency, reduce costs, and promote sustainable practices in automated logistics.

FUTURE ACTION

- Planning to submit it to a conference
- Future work may consist of extending the problem to multi agents/AGVs

REFERENCE



[1] Sperling, Marvin, and Kai Furmans. "Energy Requirement Modeling for Automated Guided Vehicles Considering Material Flow and Layout Data." Designs 8, no. 3 (2024): 48.

[2] Zhan, Xiangnan, Liyun Xu, Jian Zhang, and Aiping Li. "Study on AGVs battery charging strategy for improving utilization." Procedia CIRP 81 (2019): 558-563.

